ADVANCED PROBLEMS FOR LECTURE 17

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

(1) Calculate the limit. (Hint: think integrals)

$$\lim_{n\to\infty} \frac{1}{n}\left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \cdots + \sqrt{1}\right)$$

(2) We see that the integral function

$$F(x) = \int_a^x f(t)dt$$

is an anti-derivative of f on $[a, b]$. Show that if $c \in [a, b]$ then

$$G(x) = \int_c^x f(t)dt$$

is also an anti-derivative of f on $[a, b]$.

(3) Are all anti-derivatives of f on $[a, b]$ of the form $G_c(x)$, where $c \in [a, b]$ and

$$G_c(x) = \int_c^x f(t)dt?$$

(4) Try proving the Fundamental Theorem of Calculus.

(5) Suppose f is continuous on $[a, b]$. Show that there is a "trivial" (i.e. where the partition we take has no chopping points, so $P = \{a, b\}$) Riemann sum that perfectly captures the integral of f over $[a, b]$. That is, find a point $c \in [a, b]$ such that

$$\int_a^b f(t)dt = f(c)(b - a).$$

Hint: Does this equality remind you of any theorems we've talked about in class?